

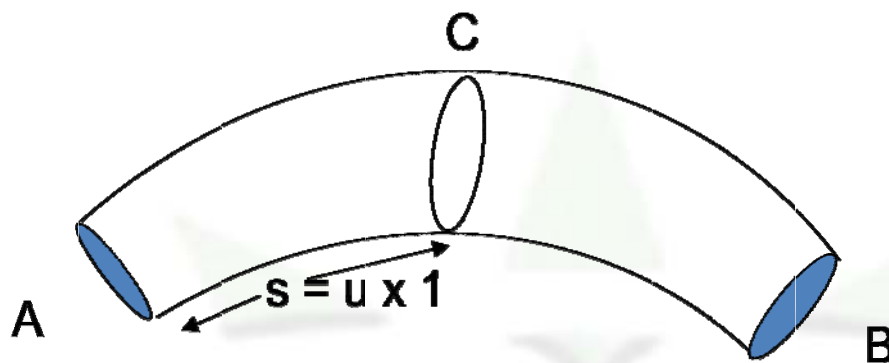


Hydrodynamics - Definition

Flow of liquid through a tube:

When a liquid flows through tube the rate of flow of liquid is defined as the liquid flowing through any section of the tube in 1 sec.

Let us consider a liquid flowing through a tube of uniform cross section.



a = area of the cross section of the tube

u = The velocity of flow of the liquid

ρ = The density of the liquid

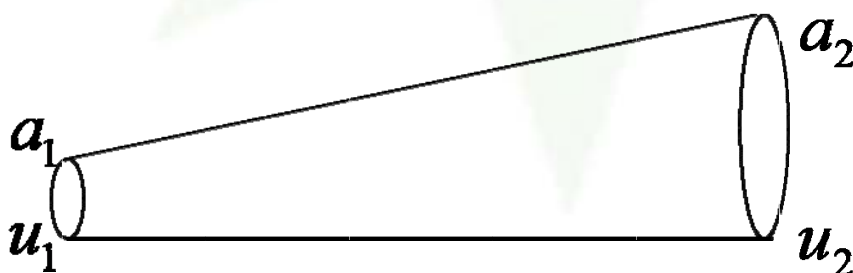
Let the liquid enters through A and go up to C in 1 sec. $AC = u \times 1 = u$

The volume of liquid flowing through the section A in 1 sec = Volume of the liquid in AC

$$= a \times u$$

$\therefore$ The mass of the liquid flowing through the section A in 1 sec

$$= a \times u \times \rho = \text{Rate of flow of liquid}$$



$$a_1 u_1 \rho = a_2 u_2 \rho$$

$$a_1 u_1 = a_2 u_2$$

If $a_1 < a_2$ then $u_1 > u_2$

The velocity of flow is greater in the region of smaller cross section & vice-versa.

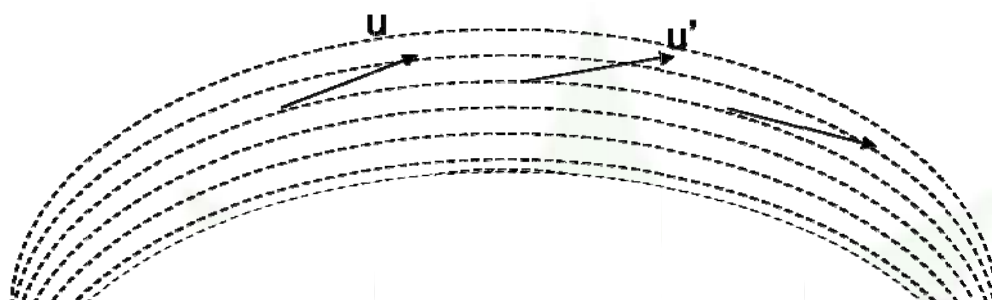


Hydrodynamics - Definition

Stream Line Motion:

When a liquid flows through a tube in such a way that every particle of the liquid exactly follows the path of its predecessor, the motion is said to be stream line.

The velocity of all the liquid particles remains same at the same point.



A stream line motion may be defined as a curve such that the tangents at any point of which gives the direction of velocity of the liquid particles at that point.

Turbulent Motion: When the pressure difference across the tube increases the velocity of flow also increases. It is found that when the velocity of flow exceeds a particular the motion loses all its steadiness, the motion becomes zigzag or sinuous or random. This motion is said to be the turbulent motion.

The velocity of flow exceeding which stream line motion becomes turbulent motion is known as **critical velocity (V_c)**.

$v < v_c \rightarrow$ stream line motion

$v > v_c \rightarrow$ *turbulent* motion

Osborn & Reynolds gave an expression for critical velocity

$$v_c = \frac{k\eta}{\rho r}$$

r = radius of the tube through which the liquid flows

ρ = density of the liquid

η = co-efficient of viscosity of the liquid

k = constant = Reynolds number.